

Welcome! We will begin shortly



Introduction to World of Data and AI

Learning Outcomes

- Summarize key Artificial Intelligence terminologies and how they are connected
- Outline the role of Generative AI in business and how it differs from other AI paradigms
- Illustrate the use of Agentic AI to solve simple point problems

Guidelines



Listen only mode



Ask questions at the interest of the larger audience



Questions in the Q&A Box

Thank you

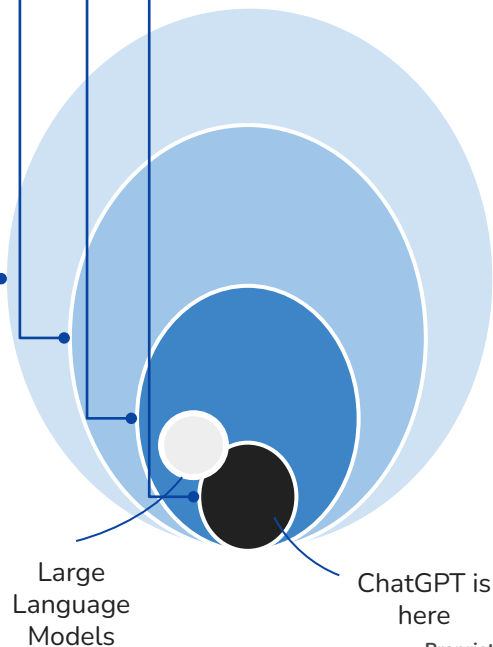
Kindly utilize the chat box for **subject-relevant questions only** to maximize your learnings from the session.

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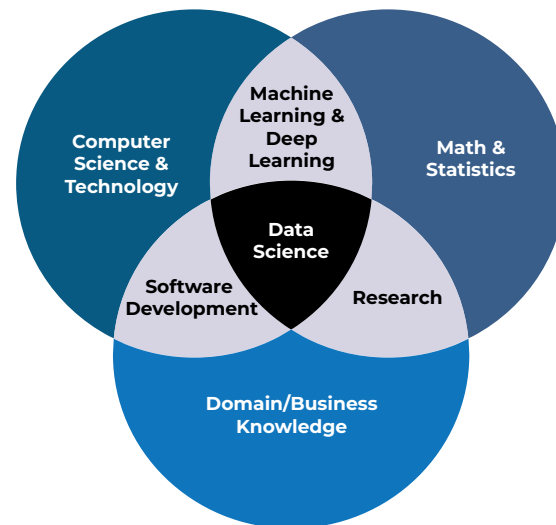
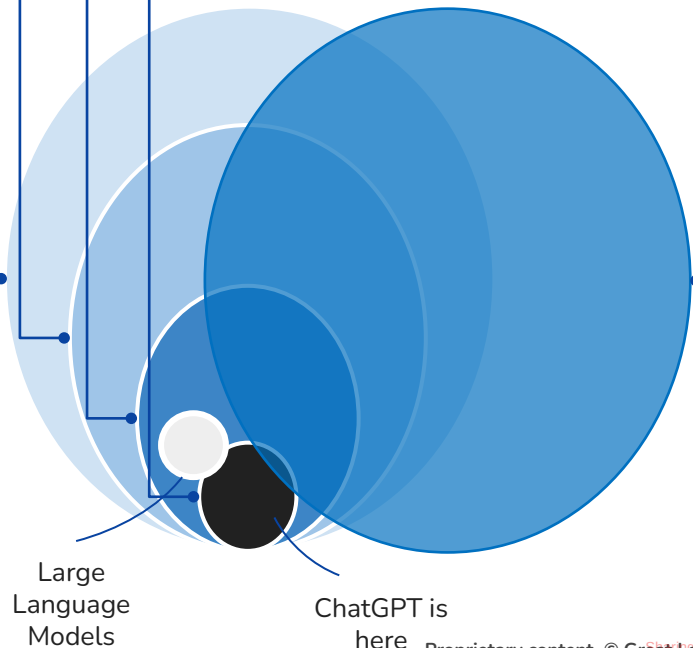
Key Terminologies in The World of Data

- **Artificial Intelligence** *Machine that mimics human intelligence & behaviour*
- **Machine Learning** *Machine that learns from the data w.r.t a particular task and performance measure*
- **Deep Learning** *Subset of ML, learning happens via algorithms inspired by neural networks in the human brain*
- **Generative AI** *Subset of DL, creating new data instances by learning the training data*

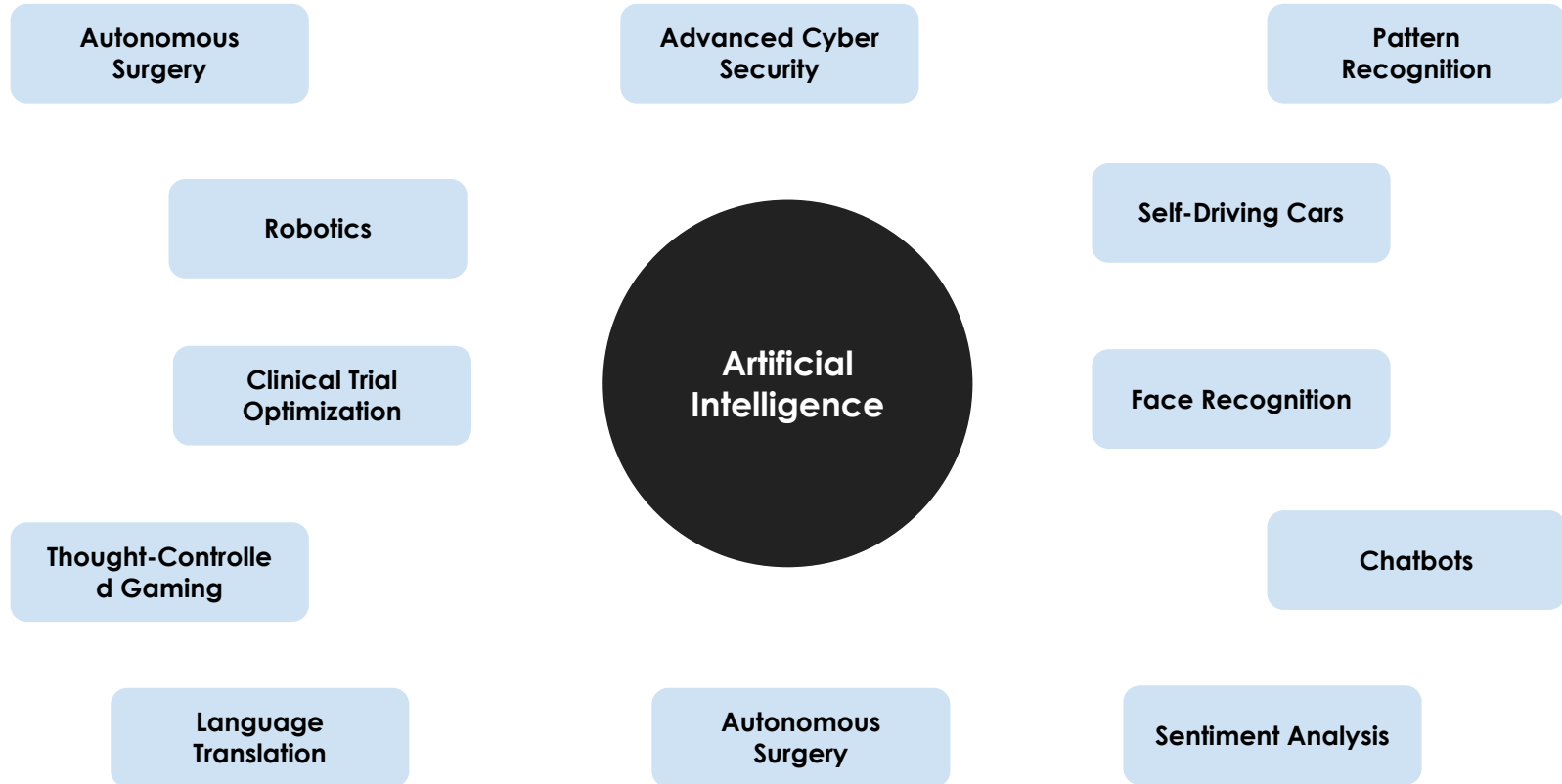


Key Terminologies in The World of Data

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- **Generative AI** *Subset of DL, creating new data instances by learning the training data*
- **Data Science** *Using AI to enable data driven decision making*

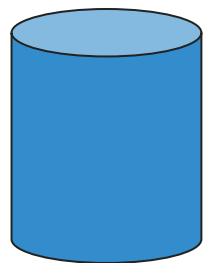


Applications of Artificial Intelligence (AI)

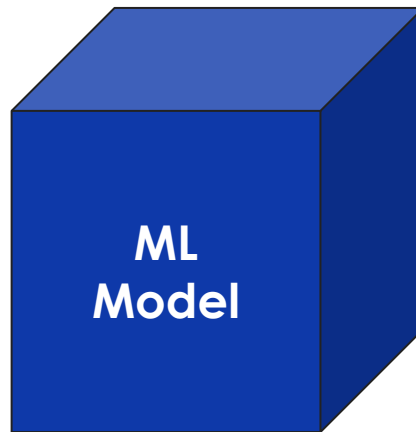


What is Machine Learning (ML)?

The model learns
the patterns in the data

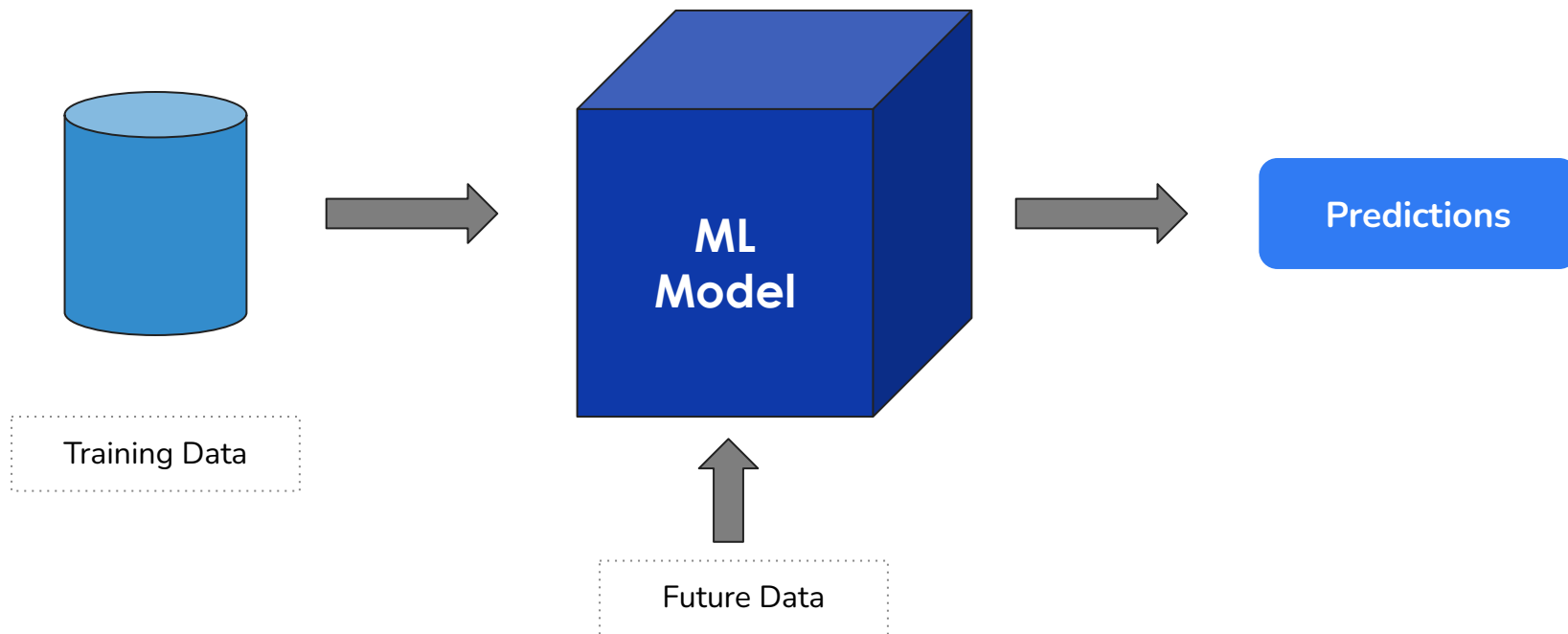


Training Data



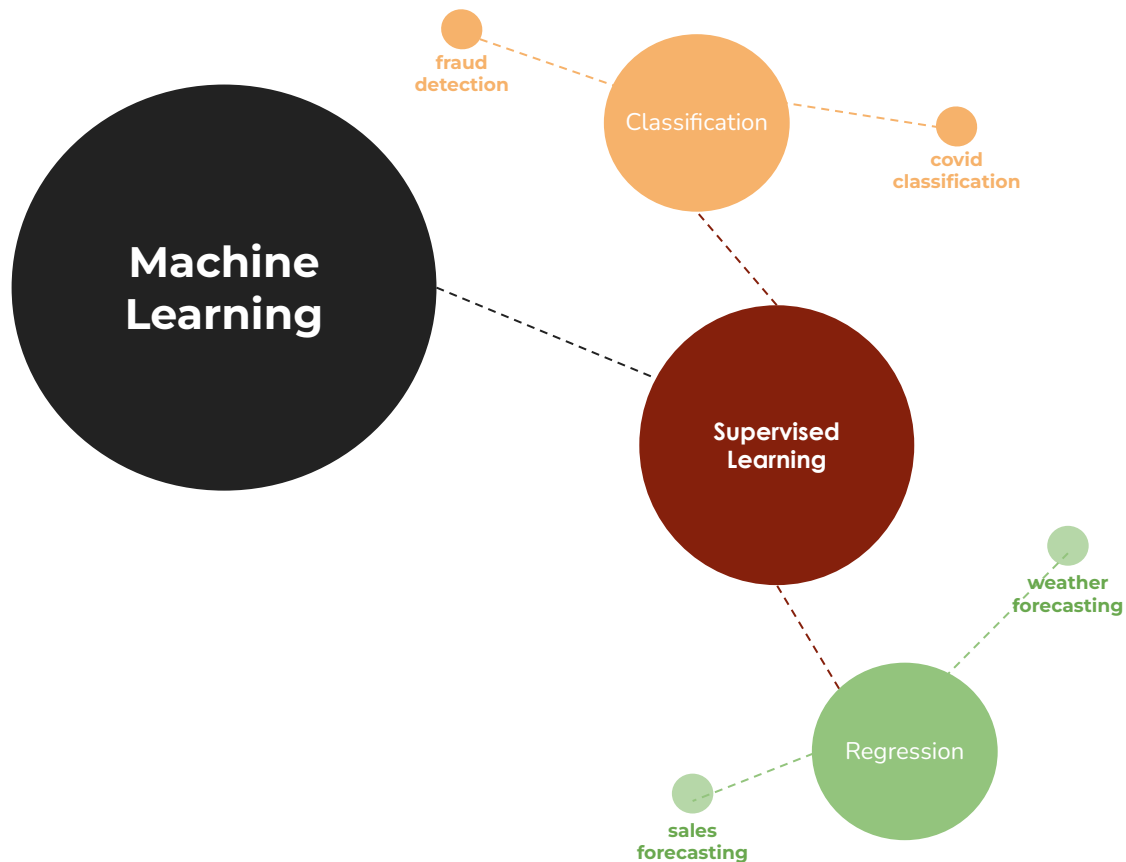
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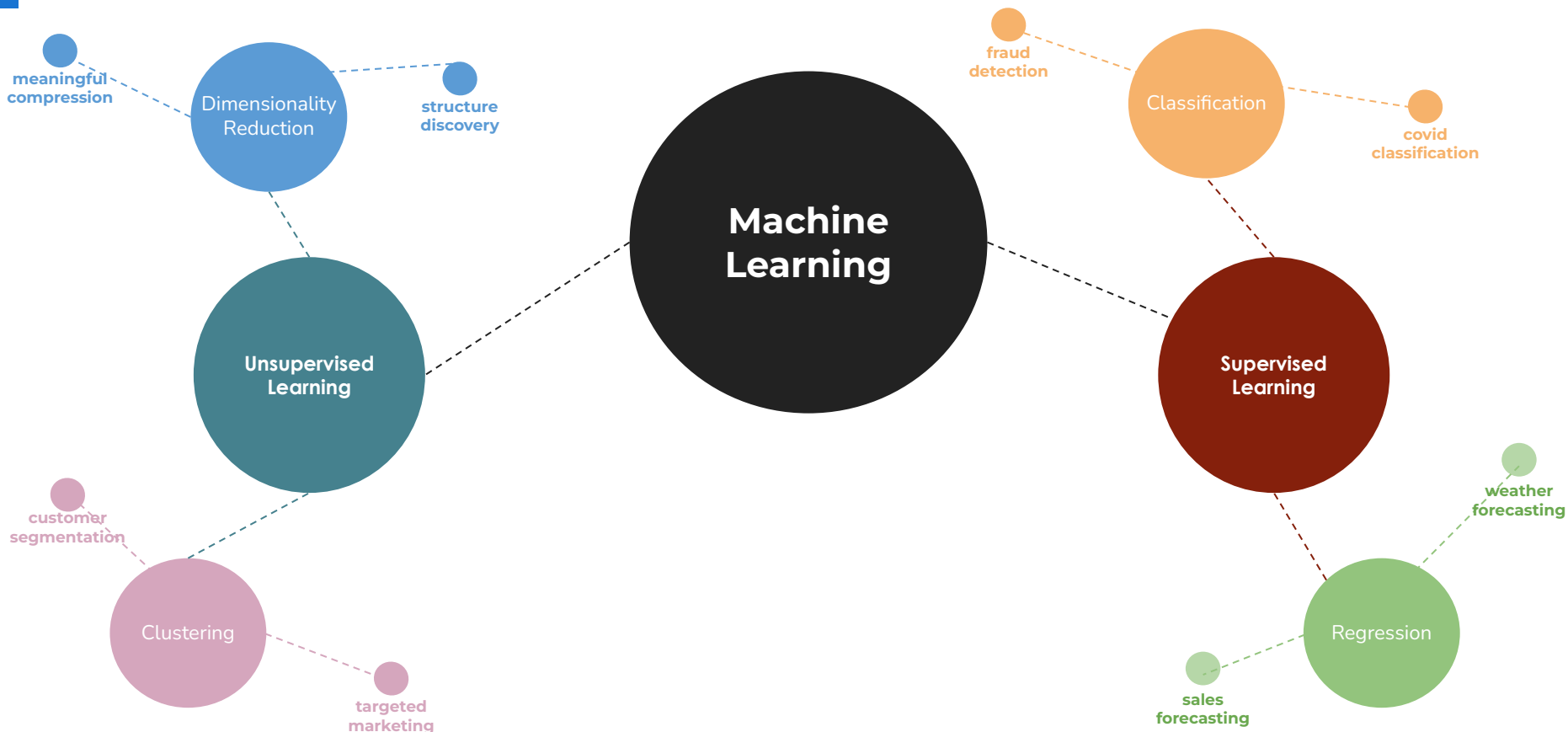


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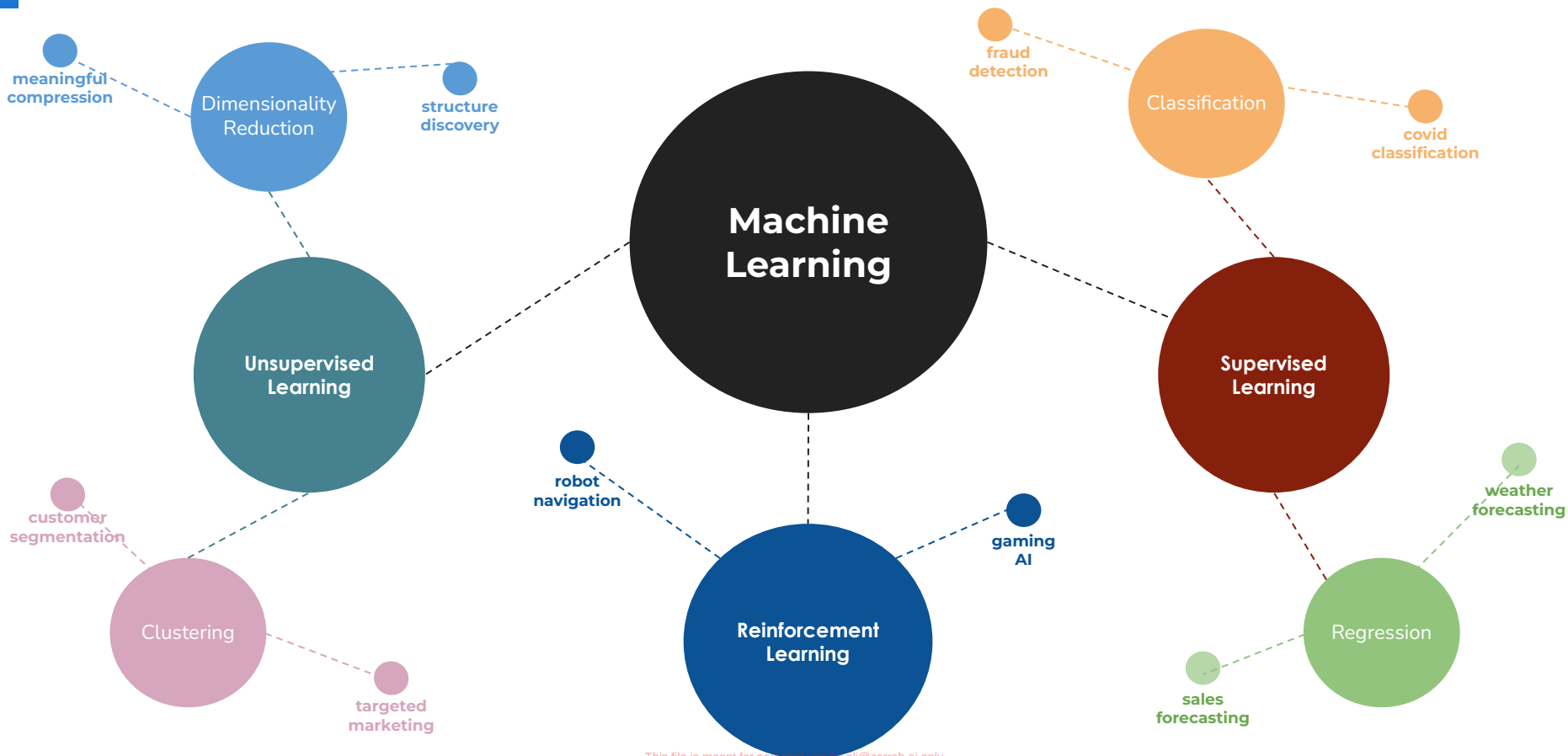
Types of Machine Learning



Types of Machine Learning



Types of Machine Learning



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Which of the following is a practical application of supervised learning?

A

Dividing the customers of an e-commerce platform into different segments

B

Visualizing high-dimensional equipment sensor data in lower dimensions

C

Predicting the price of a used car based on the attributes of the car

D

Predicting the likelihood of a hotel reservation getting cancelled

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Predicting the likelihood of a hotel reservation getting cancelled

Data Quiz

Dividing the customers of an e-commerce platform into different segments

Customers can be segmented into different categories based on their purchase and demographic attributes using unsupervised learning techniques like clustering algorithms.

Visualizing high-dimensional equipment sensor data in lower dimensions

High-dimensional data can be efficiently brought down to lower dimensions (2 or 3) for visualization purposes using unsupervised learning techniques, like PCA and t-SNE, while retaining the most important information.

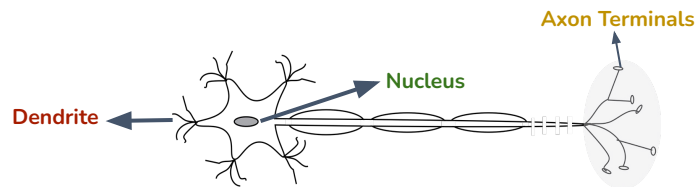
Predicting the price of a used car based on the attributes of the car

Algorithms from a subset of supervised learning, called regression, can be trained using historical data containing attributes like mileage, horsepower, manufacture year, distance driven, and more to determine the price of a used car

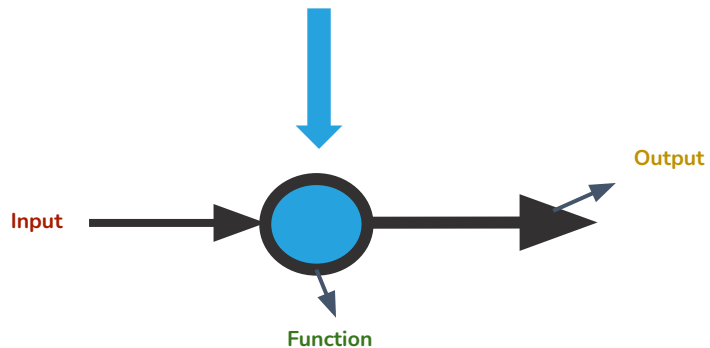
Predicting the likelihood of a hotel reservation getting cancelled

Algorithms from a subset of supervised learning, called classification, can be trained using historical data containing attributes like reservation lead time, room price, no. of guests, and more to determine the likelihood of cancellation

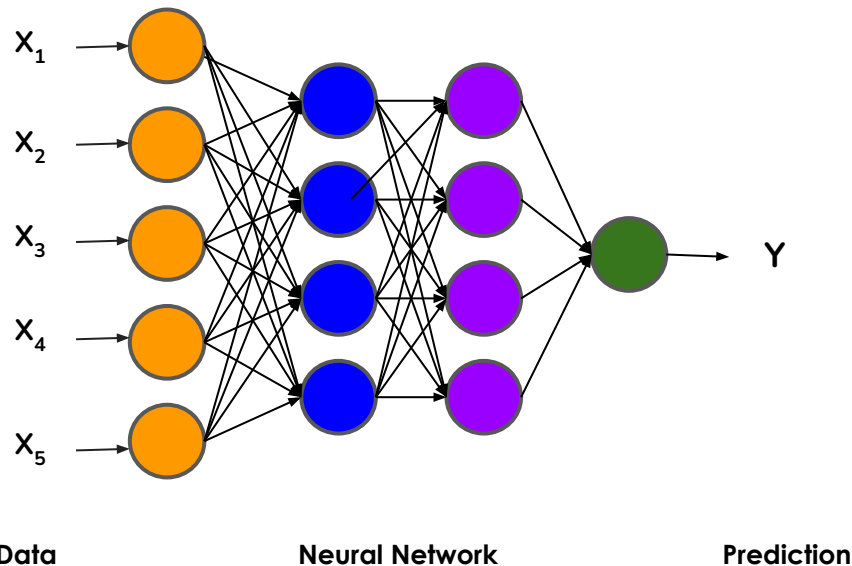
What is Deep Learning (DL)?



Biological Neuron

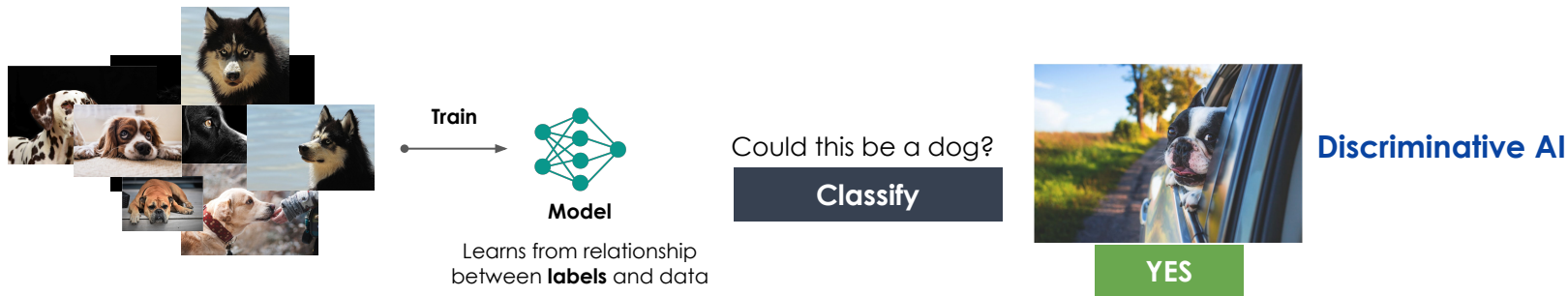


Artificial Neuron

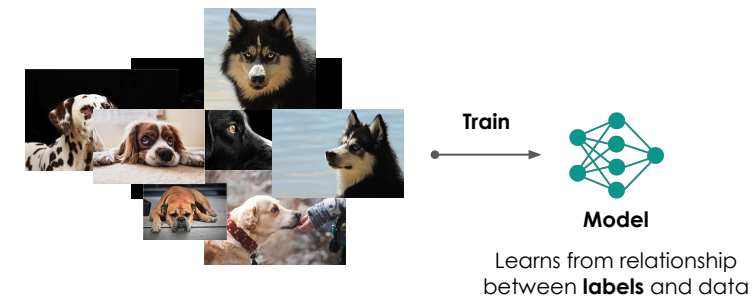


Artificial Neural Network

Types of AI

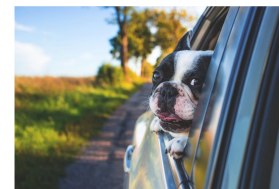


Types of AI



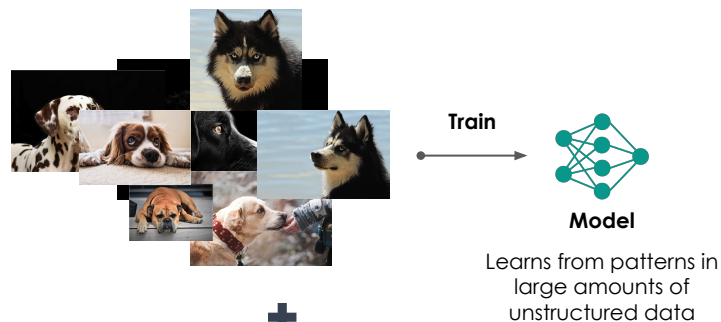
Could this be a dog?

Classify



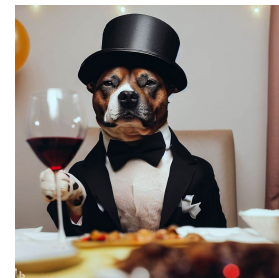
YES

Discriminative AI



Create an image of dog, having fun in a party, wearing a black tuxedo with wine in one hand

Generate



Generative AI

Other images on the internet

Which of the following is an example of a generative AI task?

A

Detecting whether an email is spam or not

B

Classifying images of animals into categories

C

Predicting whether a customer will buy a product or not

D

Creating a poem based on a topic given by the user

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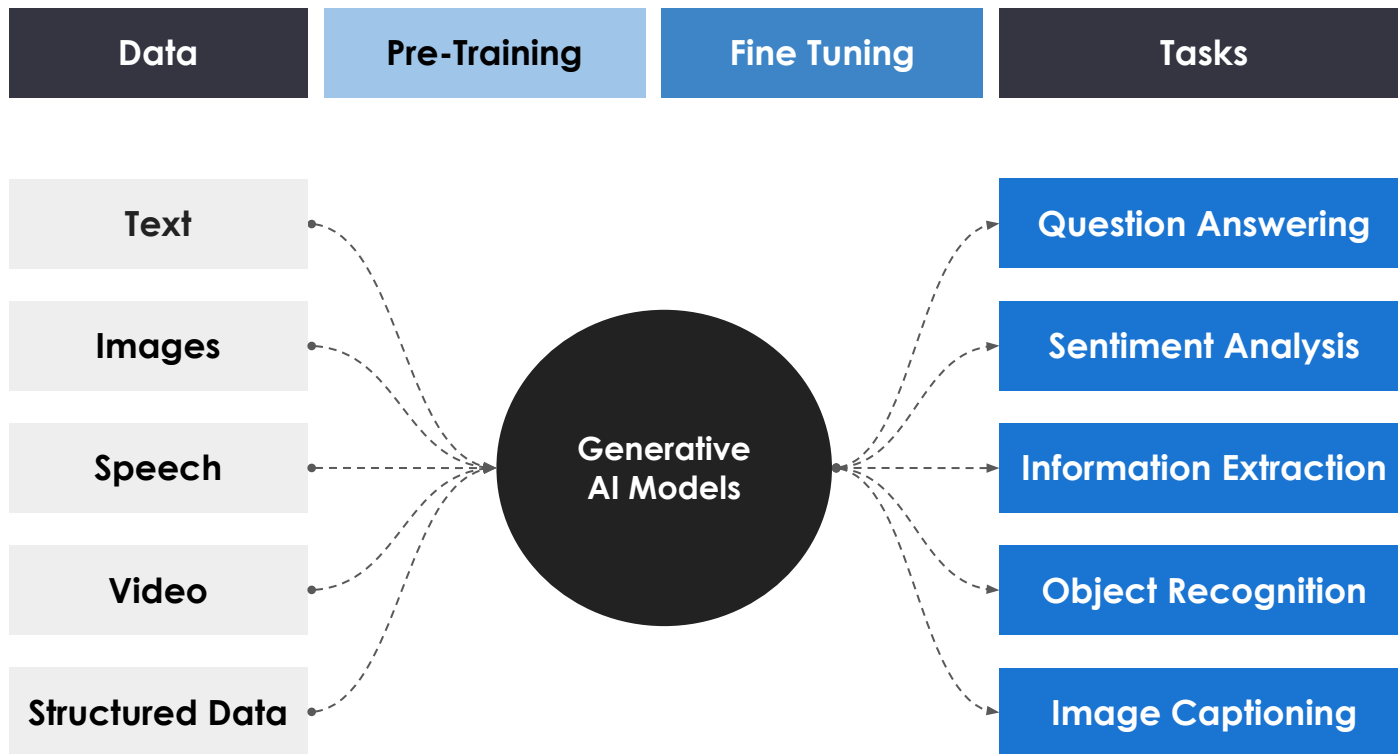
C

Predicting whether a customer will buy a product or not

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Creating a poem based on a topic given by the user

A Peek into Generative AI Models



Large Language Models

Powerful AI models trained on massive amounts of data to learn the complex patterns and rules of human language, allowing them to perform a wide variety of tasks

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Powerful AI models trained on massive amounts of data to learn the complex patterns and rules of human language, allowing them to perform a wide variety of tasks

Large

Trained on **large amounts of data** and have **billions of trainable parameters**

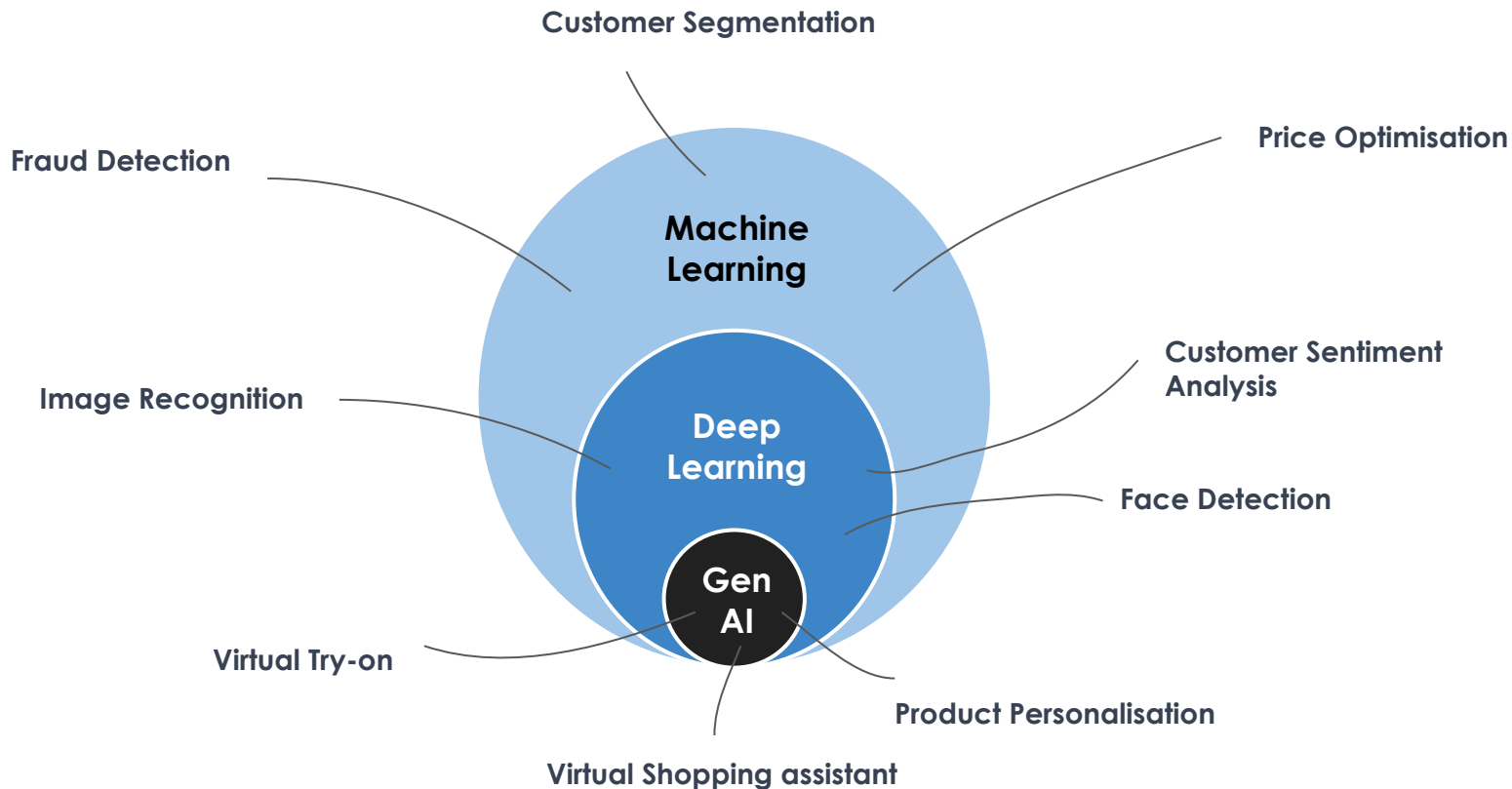
Language

Deals with text data (takes input in text and generates output in text)

Model

Predicts the next **token**

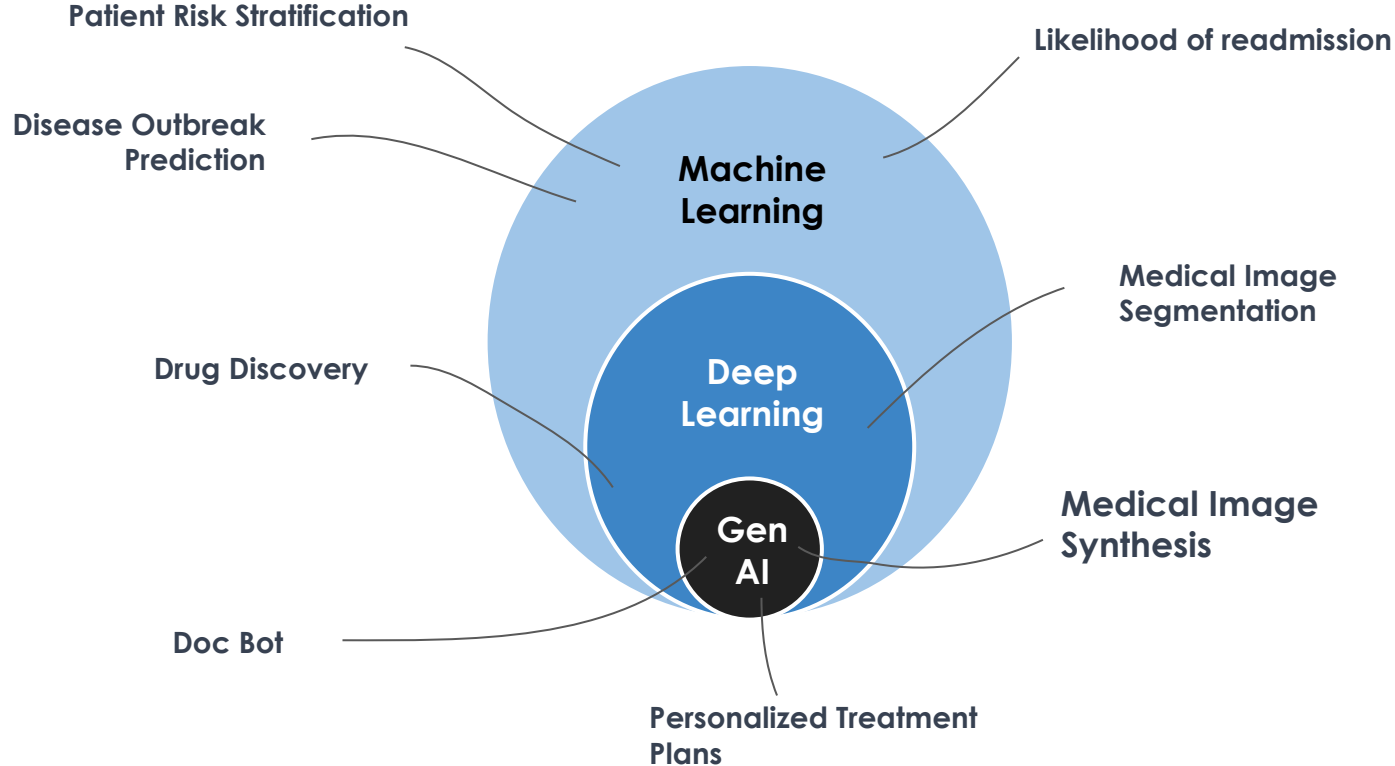
Business Problems Solved by Generative AI - Retail



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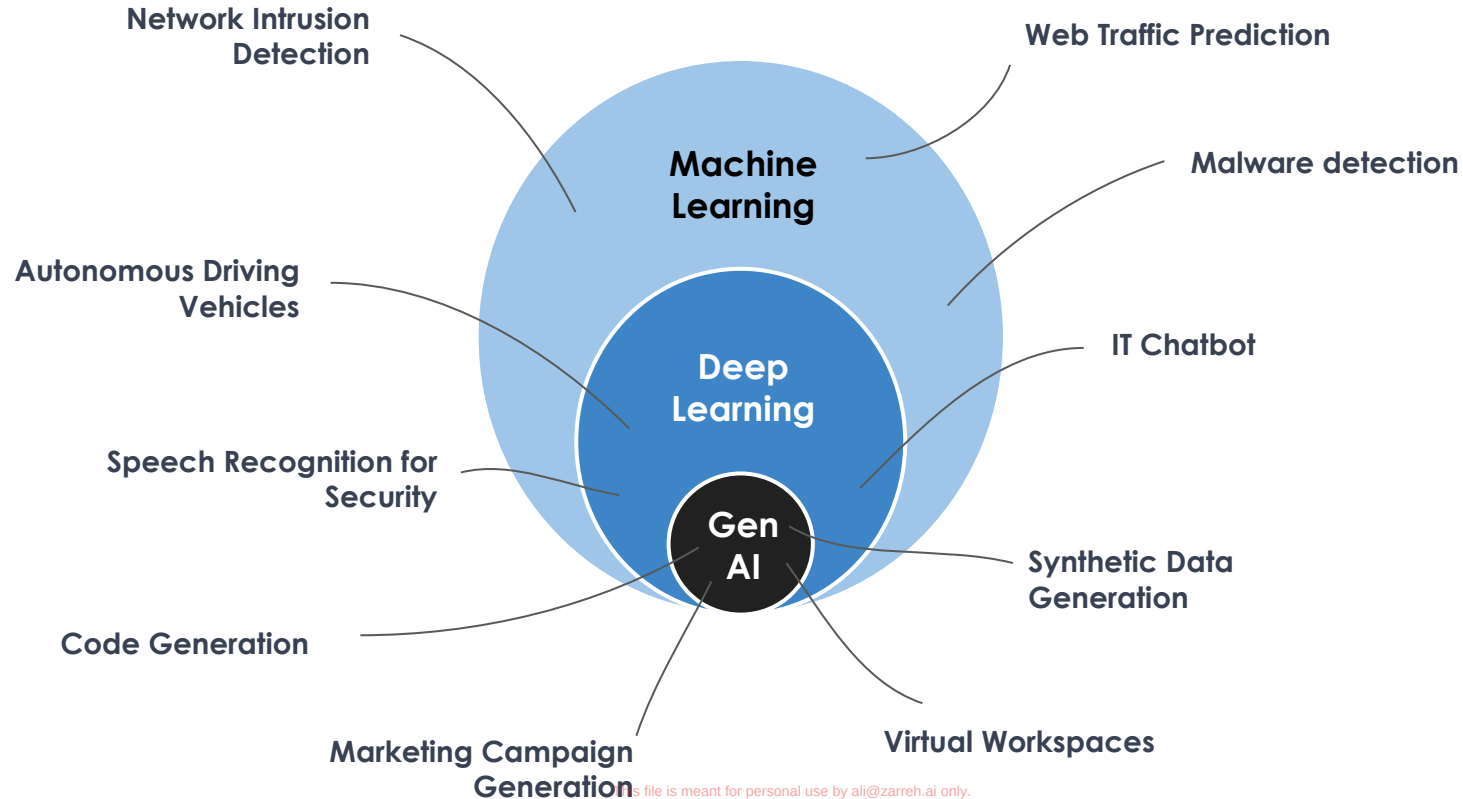
Business Problems Solved by Generative AI - Health care



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Business Problems Solved by Generative AI - Technology



Which of the following is LEAST LIKELY to be the output of a generative AI model?

A

A presentation for a lecture on data science based on provided instructions regarding content, audience, and session duration

B

Answers to questions regarding a popular tourist spot based on an images of the spot

C

A portrait of the Leaning Tower of Pisa with artistic features of a Leonardo da Vinci painting

D

Identifying whether a customer is likely to default on loan repayment based on credit history and other financial attributes

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Identifying whether a customer is likely to default on loan repayment based on credit history and other financial attributes

Data Quiz

A presentation for a lecture on data science based on provided instructions regarding content, audience, and session duration

Answers to questions regarding a popular tourist spot based on an images of the spot

A portrait of the Leaning Tower of Pisa with artistic features of a Leonardo da Vinci painting

Identifying whether a customer is likely to default on loan repayment based on credit history and other financial attributes

Generative AI models understand the relationship between instructions and presentation content by first learning from a collection of existing presentations and their instructions. They then 'decode' new instructions to generate relevant content.

The model first analyzes the visual features of the image (computer vision), then understands the question (natural language processing), and then generates a relevant answer based on its understanding of the image and the question

Generative AI models can transfer artistic styles between different image by identifying the underlying patterns. These style transfer abilities allow for the transformation of an image to adopt the artistic style of another image.

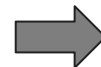
Problems like the prediction of likelihood of loan repayment default are effectively dealt with by discriminative AI algorithms. They are generally more robust in these type of tasks as they learn the decision boundaries directly.

Limitation of Large Language Models

I want to book a flight from Seattle to Portland for a meeting on June 18. What are the options available?



Large Language Model



Limitation of Large Language Models

I want to book a flight from Seattle to Portland for a meeting on June 18. What are the options available?



Large Language Model



I don't have the ability to search for real-time flight information or book flights directly. However, I can suggest some tips on how to find the best options for your flight from Seattle to Portland:

1. Flight comparison websites like Google Flights allow you to compare prices from different airlines. Set your departure city (Seattle) and arrival city (Portland) along with the date.
2. Then, from the available options, choose the flight as per your preference.

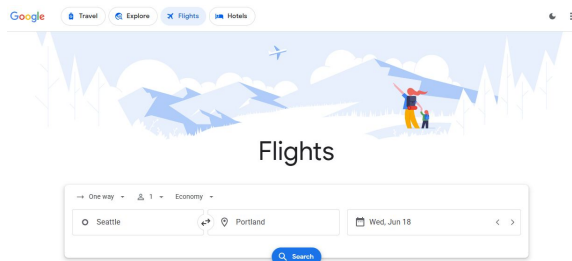
Limitation of Large Language Models

Check User Preferences

Airline Preference	No strong preference
Departure Time	Flexible
Duration	Shorter flights preferred
Stops	Nonstop preferred

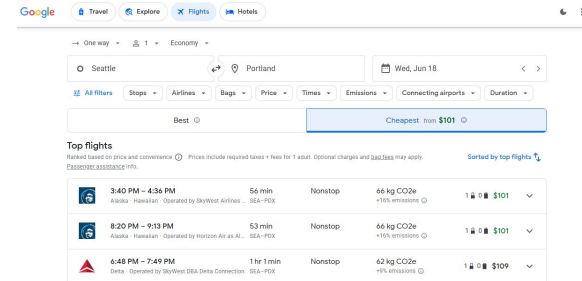
An LLM **CANNOT** identify user preferences based on past interactions

Search Flights



An LLM **CANNOT** open a browser and search for flights - it only takes text as input and produces text as output

Suggest Flights



But LLMs are good at providing steps to search for the best flight as per user preferences

Introduction to Memory

If an LLM is provided with memory, it can retain past interactions or important information and use them later to generate more relevant and contextual responses, rather than treating every request as entirely new.

Preferences of this user for flight bookings



Memory



Airline Preference	No strong preference
Departure Time	Flexible
Duration	Shorter flights preferred
Stops	Nonstop preferred
Price Sensitivity	Prefers lower prices
Seat Preference	Window or aisle
Baggage	None
Flexibility	None

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Introduction to Tools

If an LLM is provided with tools, it can use external capabilities to perform tasks beyond text generation, such as retrieving real-time information, executing code, or analyzing documents.

List of flights from Seattle to Portland on June 18



Google Search Tool

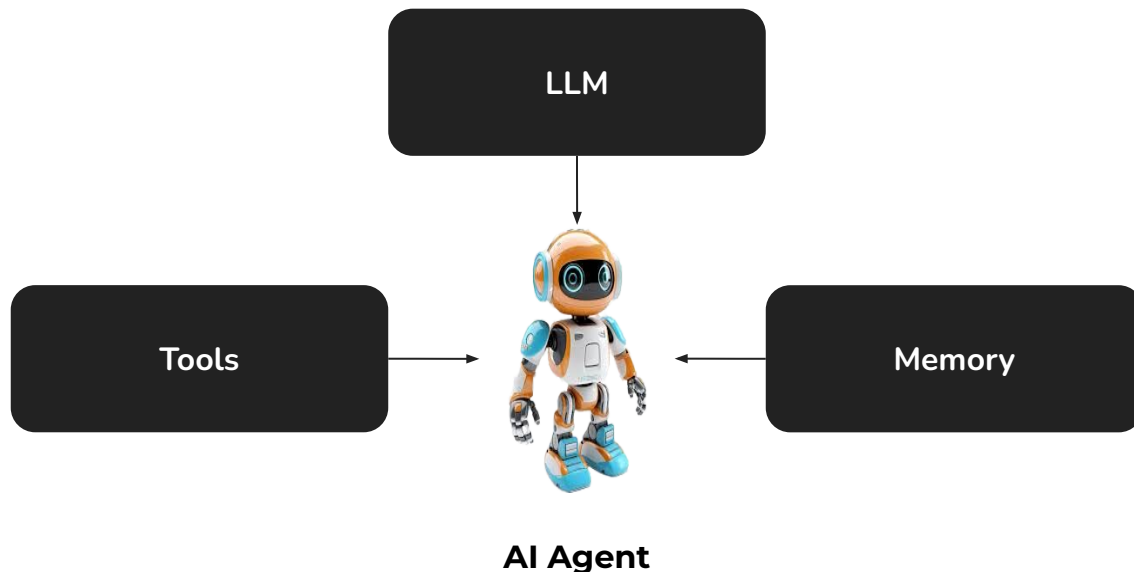


Alaska | 3:40 PM - 4:36 PM | 56 m | Nonstop | \$101
Alaska | 8:20 PM - 9:13 PM | 53 m | Nonstop | \$121
Delta | 6:48 PM - 7:49 PM | 1h 1m | Nonstop | \$109
Alaska | 12:59 PM - 1:47 PM | 48 m | Nonstop | \$131
Southwest | 8:50 AM - 5:25 PM | 8h 35 m | 1 stop (LAS) | \$152
Delta | 4:05 PM - 5:10 PM | 1h 5 m | Nonstop | \$189

Introduction to Agent

An AI Agent is an autonomous entity that perceives its environment and takes actions to achieve goals

An LLM-powered AI agent follows the same idea. It perceives inputs such as user queries and context, applies reasoning to identify what needs to be done, and takes actions by using tools and memory to achieve the goal.



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Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



First, I'll fetch the user's past preferences to understand their needs.

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Let me use my memory to get the details

Memory

Airline Preference | No strong preference
Departure Time | Flexible
Duration | Shorter flights preferred
Stops | Nonstop preferred
Price Sensitivity | Prefers lower prices

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Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Now, I'll send the results back to the LLM

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



LLM

Airline Preference | No strong preference
Departure Time | Flexible
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Price Sensitivity | Prefers lower prices

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Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



LLM

Based on your past preferences, you don't have a strong airline preference, your departure time is flexible, you prefer shorter and nonstop flights, and you tend to favor lower-priced options. Should I proceed with these, or would you like to make any changes?

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Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Now, I'll confirm the user's past preferences for the current booking.

Agent in Action!

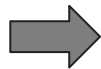
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Based on your past preferences, you don't have a strong airline preference, your departure time is flexible, you prefer shorter and nonstop flights, and you tend to favor lower-priced options. Should I proceed with these, or would you like to make any changes?

Agent in Action!

Yes, that looks good. Please proceed with these preferences.



Agent in Action!

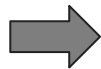
Yes, that looks good. Please proceed with these preferences.



Now, I'll send the response to the LLM

Agent in Action!

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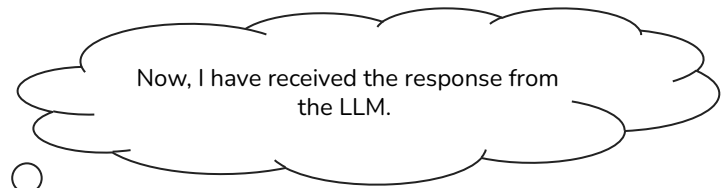
Now, I'll send the results back to the LLM

LLM

Yes, that looks good. Please proceed with these preferences.

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



LLM

Invoke `Google_Search_Tool`

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Now, I'll use the Google Search tool to find flight options based on the user's preferences.

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Now, I'll use the Google Search tool to find flight options based on the user's preferences.

Google Search
Tool

Alaska | 3:40 PM - 4:36 PM | 56 m
| Nonstop | \$101
Alaska | 8:20 PM - 9:13 PM | 53 m
| Nonstop | \$121
Delta | 6:48 PM - 7:49 PM | 1h 1m
| Nonstop | \$109
Alaska | 12:59 PM - 1:47 PM | 48
m | Nonstop | \$131

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Now, I'll send the results back to the LLM

Agent in Action!

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LLM

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| Nonstop | \$131



Now, I'll send the results back to the LLM

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Now, I have received the response from the LLM.

LLM

Here are the best options based on your preferences:

Alaska: 3:40 PM – 4:36 PM (56m, Nonstop) – \$101
Alaska: 8:20 PM – 9:13 PM (53m, Nonstop) – \$121
Delta: 6:48 PM – 7:49 PM (1h 1m, Nonstop) – \$109
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The cheapest option is \$101, and the shortest flight is 48 minutes. Which one would you like to choose?

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Agent in Action!

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Here are the best options based on your preferences:

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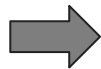
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Alaska: 12:59 PM – 1:47 PM (48m, Nonstop) – \$131

The cheapest option is \$101, and the shortest flight is 48 minutes. Which one would you like to choose?

Agent in Action!

I'll go with the Alaska flight at 3:40 PM.



Agent in Action!

I'll go with the Alaska flight at 3:40 PM.



Now, I'll send the response back to the LLM

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Now, I'll send the response back to the LLM

LLM

I'll go with the Alaska flight at 3:40 PM.

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?

LLM

The user's preferred flight from Seattle to Portland on June 18 is with Alaska: 3:40 PM - 4:36 PM, 56m, Nonstop, priced at \$101.



Now, I have received the response back from the LLM

Agent in Action!

I want to book a flight from Seattle to Portland for a meeting on June 18. I'm open to any time. What are the options available?



Your preferred flight from Seattle to Portland on June 18 is with Alaska: 3:40 PM - 4:36 PM, 56m, Nonstop, priced at \$101.

Agent in Action!



Stopping the process now!

Interacting with Generative AI Models

Hands-on: Exploring Agentic AI Tools



Thank you!

We'd love to hear your feedback!
Please share your feedback for the session

Wish you all the very best!

Please feel free to raise a [Support Request](#) through Olympus in case of any queries